

K1. Clear and Comprehensive Summary

A. Defining the Problem (Elements A-C)

The primary problem identified was the lack of an affordable, accessible, and reliable method for measuring maximal inspiratory pressure (MIP) in a classroom setting with a large cohort of students. Existing solutions, such as commercially available MIP devices, were found to be too expensive for large-scale implementation, often exceeding several hundred dollars per device (up to \$2000). Additionally, these devices were not always optimized for ease of use in an educational setting, limiting their practicality for student training.

To address this, key design requirements were established, including durability (lasting at least five years), accuracy (within ± 1 inHg), simplicity and replicability (capable of being produced in quantities of 10-20 units), affordability (low-cost materials and construction), and portability (compact and easy to store for classroom use).

B. Ideation, Selection & Prototype Design (Elements D-F)

Several design concepts were explored, including mechanical pressure gauges, digital sensors, and water based designs. After evaluating feasibility, cost, and ease of replication, the final design was selected based on its ability to meet the high-priority requirements while keeping the cost low. The chosen design incorporated a mechanical pressure gauge housed in a 3D-printed casing, eliminating the need for complex electronics that would increase cost and maintenance requirements.

The initial prototype was planned with a pressure gauge for measurement, a mouthpiece for ease of use, and a 3D-printed tube to connect the gauge to the mouthpiece. The design focused on minimizing material costs while ensuring accuracy through proper sealing and airflow management.

C. Prototype Building (Element G)

The prototype was constructed using PLA filament for the 3D-printed casing, a standard pressure gauge calibrated for respiratory measurements, and silicone tubing for airflow regulation. One of the challenges faced during construction was ensuring an airtight seal between components to prevent measurement inaccuracies.

A major design choice justified using expert feedback was selecting a mechanical pressure gauge over an electronic sensor. Experts in biomedical engineering advised that while electronic sensors could offer digital readouts, they introduced potential calibration issues and cost barriers, making them impractical for the intended classroom application.

D. Testing & Evaluation (Elements H-J)

The main testing goals included validating the accuracy of pressure readings, assessing the durability of the 3D-printed casing, and ensuring ease of use for students and instructors. Testing was conducted by performing repeated trials, comparing results to commercially available MIP devices, and analyzing feedback from field experts and stakeholders.

Expert and stakeholder feedback indicated that the device performed well in terms of accuracy but required minor design modifications to improve user comfort and sealing effectiveness.

Additionally, durability testing revealed that thicker casing walls improved structural integrity without significantly increasing material costs.

K2. Value Judgment on the Process

A. What Went Well

There were several things that went quite well during our project. The part of the project that went the best was probably the portability. The constraints regarding portability were relatively lax and didn't require a lot of alterations or adjustments. We are quite certain that our device will be able to be transported and carried easily. Another thing that went well is the affordability. It costs around \$15 to make one of these devices, and considering our initial budget estimate of \$30-40 we exceeded all expectations.

B. What Didn't Go Well

A challenge encountered during the project was time management. We wasted lots of time that would have been spent on valuable work if we had gotten things right the first time, set realistic deadlines and goals, and removed distractions. Another challenge was figuring out which design was best for the mouthpiece. This includes different sizes, and shapes. We wanted something that would be efficient, safe, and easy to clean.

C. Why?

The primary reason for the success of keeping costs low was careful material selection and eliminating unnecessary electronic components. The main reason we struggled with focusing is

the lack of time constraints leading to us making the conscious decision to not work and instead talk to others or play games. If given another opportunity, I would improve the prototype by exploring alternative methods for measuring such as a spring based design to improve precision.

K3. Lessons Learned

A. The Importance of a Structured Design Process

One lesson learned about defining design requirements is that prioritizing essential features early on helps prevent unnecessary complexity. A key takeaway from comparing multiple designs is that even seemingly minor differences in material selection and construction can significantly impact performance, cost, and usability.

B. Effective Testing & Evaluation Strategies

One major insight from testing is that repeated trials with diverse users are essential for identifying inconsistencies and refining the design for broader usability. The importance of real-world testing was highlighted by how initial assumptions about airflow sealing needed validation through hands-on use, leading to necessary design adjustments.

C. The Role of Expert & Stakeholder Feedback

Expert feedback was particularly useful for determining the most practical measurement method, as early considerations included digital and electronic options that ultimately proved unnecessary. Engaging stakeholders throughout the process helped in ensuring that the final design aligned with user needs, particularly regarding ease of use in a classroom setting.

D. Managing Project Constraints

A challenge related to budget/time was balancing affordability with durability, which was addressed by optimizing 3D print settings and selecting cost-effective yet sturdy materials. The most effective strategy for balancing constraints was prioritizing high-impact design decisions first, such as focusing on accuracy and durability, while leaving aesthetic considerations as a lower priority. Similarly, another lesson learned is that time management in relation to the time that's needed to build and test the device is very important.